



**Verified Carbon
Standard**

FX GROUP BIOMASS TO ENERGY PROJECT



BRUNDTLAND

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Prepared by	Brundtland (Pty) Ltd.

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1 PROJECT DETAILS

1.1 Summary Description of the Project

The FX Group Biomass to Energy Project (hereinafter, the Project) is developed by FX Group (Pty) Ltd. (hereinafter, the Project Proponent) and is a woody biomass residue utilization project. The proposed project activity concerns a greenfield biomass to energy power plant at the FX Group Lothair facility, which is situated in the Mpumalanga province of South Africa, in the eastern region of the country, bordering the nations of Mozambique and Eswatini. The facility manufactures high-quality engineered wood panel products.

The Project will source biomass from local sawmills and biomass wastes from the FX Group's Lothair operation for the generation of electricity in a power-only plant. The woody biomass will be fed into a biomass boiler to create super-heated high-pressure steam. The steam is used to spin a 4.5MW power output steam turbine and electrical generator. The proposed project activity provides renewable energy to FX Group's Lothair industrial facility, with an estimated 20 463 MWh generation per year.

The renewable energy produced by the plant will displace grid electricity, which would have been used to power the industrial facility in the absence of the Project. Electricity on the South African National grid is largely generated by the state-owned entity Eskom and is highly carbon-intensive due to the reliance on fossil-fuels such as coal and diesel. The Project is expected to provide average GHG reductions of 17 221 tCO₂e per year.

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
Validation/ verification	Validation site visit	VCS	TUV-Nord	
	02 - July -2024			
	– 05 – July - 2024			

1.3 Sectoral Scope and Project Type

Sectoral scope ¹	1. Energy (renewable/non-renewable)
Project activity type	Stand-alone

¹ Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: <https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes>

1.4 Project Eligibility

1.4.1 General eligibility

The Project is eligible to participate in the VCS program because:

- The project generates renewable electricity from biomass residues waste streams that would have been discarded and landfilled in the absence of the Project. The project activity is not excluded under table 2.1 of the VCS standard (version 4.7)
- The Project documentation was uploaded onto the VCS website for pipeline listing on 27 May 2024 and for validation listing on 17 June 2024. The validation opening meeting with the VVB will be held on the 2nd of July. The Project meets the pipeline listing and validation listing requirements
- The Project applies methodology 'AMS-I.F Renewable Electricity Generation for captive use and mini-grid v5.0'. The methodology is classified as “active” under the CDM and VCS program. The Project has an installed capacity of 4.5MW which falls within the capacity limit of 15MW (small-scale threshold) set for the methodology. The Project is a greenfield development and consists of one electricity generator. All generation capacity is included in the Project activity and the Project is therefore not part of a fragmented project activity. The Project meets the methodology eligibility criteria

1.4.2 AFOLU project eligibility

Section not applicable as this is not an AFOLU project.

1.4.3 Transfer project eligibility

Section not applicable as this is not a transfer project.

1.5 Project Design

- ☒ Single location or installation
- ☐ Multiple locations or project activity instances (but not a grouped project)
- ☐ Grouped project

1.5.1 Grouped project design

Section not applicable, as this is not a grouped project.

1.6 Project Proponent

Organization name	FX Group (Pty) Ltd.
Contact person	Mr. Roddy Payne
Title	General Manager
Address	10 Bradford St, Benoni South, Benoni, East Rand, Gauteng, 1501, South Africa
Telephone	+ 27 11 422 4839 / +27 71 611 8168
Email	rpayne@thefxgroup.co.za

and

Organization name	Brundtland (Pty) Ltd.
Role in the project	Consultant / Carbon Aggregator
Contact person	Mr. Henk Sa
Title	Director
Address	Firestation, 16 Baker Street, Rosebank, 2196, Gauteng, South Africa
Telephone	+27 11 447 7892 / +27 78 459 5748
Email	Henk.Sa@Brundtland.co.za

1.7 Other Entities Involved in the Project

No other entities are involved in the project.

1.8 Ownership

FX Group (Pty) Ltd. is the project proponent in that it has overall control and responsibility for the Project. It has the sole right to the GHG emission reductions or removals for the project crediting period.

Project ownership arises according to property or contractual right in the plant, equipment and process that generates GHG emission reductions, as per the VCS Standard v4.7 section 3.7, paragraph 3.7.1, evidence type 3. Project ownership is demonstrated by unit handover documentation from the EPC contractor, which supports the property and contractual ownership of the plant by the project proponent.

1.9 Project Start Date

Project start date	12-Oct-2022
Justification	The project start date is the commissioning date of the biomass power generation plant, which is the date of the first GHG emission reduction.

1.10 Project Crediting Period

Crediting period	☐ Seven years, twice renewable ☒ Ten years, fixed ☐ Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)
Start and end date of first or fixed crediting period	12-Oct-2022 to 11-Oct-2032

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

- ☒ < 300,000 tCO₂e/year (project)
☐ ≥ 300,000 tCO₂e/year (large project)

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO ₂ e)
12-Oct-2022 to 31-Dec-2022	3 842
01-Jan-2023 to 31-Dec-2023	17 221
01-Jan-2024 to 31-Dec-2024	17 221
01-Jan-2025 to 31-Dec-2025	17 221
01-Jan-2026 to 31-Dec-2026	17 221
01-Jan-2027 to 31-Dec-2027	17 221
01-Jan-2028 to 31-Dec-2028	17 221
01-Jan-2029 to 31-Dec-2029	17 221
01-Jan-2030 to 31-Dec-2030	17 221

01-Jan-2031 to 11-Oct-2031	17 221
01-Jan-2032 to 11-Oct-2032	13 352
Total estimated ERRs during the first or fixed crediting period	172 233
Total number of years	10
Average annual ERRs	17 221

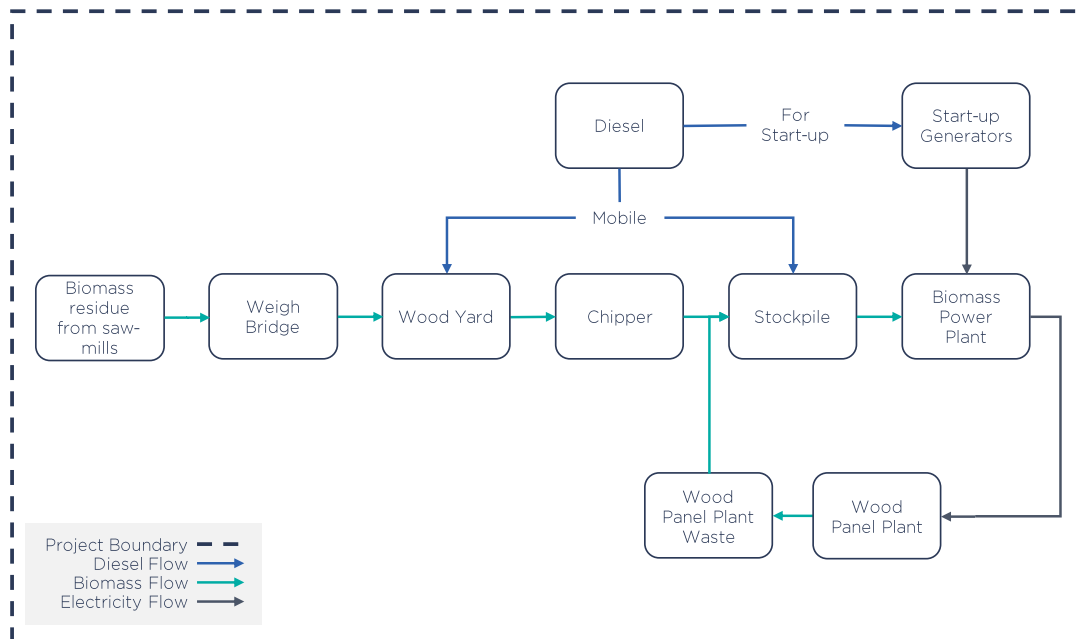
1.12 Description of the Project Activity

The project activity consists of a 4.5 MW greenfield biomass-to-energy power plant at the FX Group Lothair facility. The objective of the power plant is to burn waste biomass residue from sawmills and the wood panel plant to generate power using a conventional steam cycle electricity generation plant, which is then used to power the Lothair wood panel plant. The power plant is designed to fulfill all the electricity needs of the wood panel manufacturing plant at any given point in time, to allow it to operate independently from the South Africa national grid. The technical specifications of the three main components of the power plant, i.e., the biomass boiler, steam turbine and electricity generator, are presented in the table below.

Biomass Boiler	Specification
Manufacturer	New Nation
Quantity	1 set
Rated evaporation capacity	22 t/h
Design pressure	4.2 MPa
Rated steam pressure	3.82 MPa
Rated steam temperature	450 °C
Design life	30 Years
Steam Turbine	Specification
Manufacturer	Hangzhou Hangfa Electrical Equipment CO., Ltd.
Model	N4.5-3.43
Quantity	1 set
Rated Output	4500 kW
Rated Steam Pressure	3.43 MPa
Rated Steam Temperature	445 °C
Rated Speed	3000 (r/min)
Design life	30 Years

Electricity Generator	Specifications
Manufacturer	Hangzhou Hangfa Electrical Equipment CO., Ltd.
Model	QF-J4.5-2
Quantity	1 set
Rated Power Capacity	5.625 MVA
Rated power	4.5 MW
Rated Voltage	11 kV
Rated Current	295.2 A
Exciting Current	241.6 A
Power factor	0.8
Frequency	50 HZ
Rated Speed	3000 (r/min)
Design life	30 Years

The biomass residues consist of eucalyptus and wattle, pine with bark and sawdust from sawmills which would otherwise have been disposed of or landfilled. Incoming biomass residue is weighed via a weighbridge, after which they are transported to a wood-yard for storage. The biomass residue, including biomass residue wastes from the wood panel plant itself, is processed using a chipper. The shredded biomass residue is subsequently moved to a stockpile after which it is fed to the boiler via a conveyer belt, where it is combusted at a rate of approximately 9 tonnes per hour. The complete process is depicted in the diagram below.



1.13 Project Location

The FX Group (Pty) Ltd. biomass-to-energy project is located within the Msukaligwa Local Municipality, 65 km northeast of Ermelo at R33, 2370, Lothair, Gert Sibande District Municipality, Mpumalanga Province, South Africa. The central coordinates of the project location are: -26.390478, 30.445225. The maps and satellite imagery below indicate the geographical location of the FX Group biomass to energy project.

South Africa



Mpumalanga Province



FX Group Biomass to Energy Plant Location



1.14 Conditions Prior to Project Initiation

The FX Group biomass to energy project is a greenfield project, which is being developed in conjunction with FX Group's Lothair integrated wood panel plant. The conditions prior to the Project activity and construction of the integrated wood panel plant comprises of an area in a natural state with no construction or industrial activity. The FX Group's Lothair integrated wood panel plant is constructed to create a particle board line, a resin manufacturing plant and a melamine upgrading area, to improve the groups operations and manufacturing capacity. The powerplant will provide for the full electricity needs of the wood panel manufacturing plant and, therefore, the Project has not been implemented to generate GHG emissions for the purpose of their subsequent reduction, removal or destruction.

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

The Project complies with all relevant law and regulations of the Republic of South Africa. Compliance is evidenced by the following:

- Environmental Impact Assessment (EIA) / Environmental Authorization (EA) granted by the Department of Forestry, Fisheries and the Environment (DFFE) – Reference number 17/2/3 GS-226 (Approved)
- Water Use License – Licence number: 05/W55B/AG/13316, Reference number: WU22918 (Approved)
- Provisional Atmospheric Emission Licence– Reference number: Msukaligwa/The FX Group (Pty) Ltd/0043/2023/F01 (Approved)

The Project thereby complies with the South African National Environmental Management Act (NEMA) (No. 107 of 1998) and the Environmental Impact Assessment Regulations (2010). The project further complies with Chapter 4 of the National Water Act (No. 36 of 1998) and the National Environmental Management: Air Quality Act (No. 39 of 2004) as amended.

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

1.16.2 Registration in Other GHG Programs

Has the project registered under any other GHG programs?

☐ Yes ☒ No

Is the project active under the other program?

☐ Yes ☒ No

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit?

☐ Yes ☒ No

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system?

☐ Yes ☒ No

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☒ Yes ☐ No

If yes:

Is the project proponent(s) or authorized representative a buyer or seller of the product(s) (goods or services) that are part of a supply chain?

☒ Yes ☐ No

If yes:

Has the project proponent(s) or authorized representative posted a public statement on their website saying, "Carbon credits may be issued through Verified Carbon Standard project [project ID] for the greenhouse gas emission reductions or removals associated with [project proponent

or authorized representative organization name(s)] [name of product(s) whose emissions footprint is changed by the project activities].”

☐ Yes

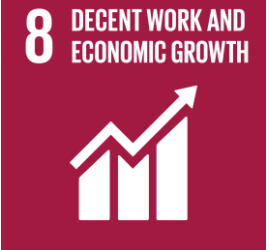
☒ No

1.18 Sustainable Development Contributions

SDG 13		
	Target 13	Take urgent action to combat climate change and its impacts.
The FX Group biomass to energy project produces renewable energy, which results in the avoided use of carbon-intensive national grid electricity, which is generated primarily from fossil fuels such as coal and diesel. Therefore, the project activity contributes to reducing the tonnes of greenhouse gas emissions emitted by fossil-fuel driven power stations to power the Lothair industrial facility. This is in line with the objectives stated in South Africa's Nationally Determined Contribution (NCD) under the Paris Agreement and Low Emissions Development Strategy (LEDS)		

SDG 7		
	Target 7	Ensure access to affordable, reliable, sustainable and modern energy for all.
The FX group biomass to energy project produces renewable energy through the use of residual biomass and a biomass boiler to create super-heated steam which is required to spin a 4.5MW steam turbine electrical generator. By creating renewable energy for consumption at the FX Group Lothair facility, the project activity contributes to increasing the renewable energy share in the total final energy consumption. This is in line with the objectives stated in South Africa's Integrated Resource Plan (2019).		

SDG 8		
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	Target 8	Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all.
The FX group biomass to energy project creates employment opportunities for the collection of biomass residue from nearby forestry plantations. FX Group purchases biomass residue from individuals who collect it from the nearby plantations and use it as fuel for their biomass boilers. By creating employment opportunities for surrounding communities, the project activity reduces the unemployment rate. This is in line with the objectives stated in the South African Constitution.		

1.19 Additional Information Relevant to the Project

1.19.1 Leakage Management

In accordance with the applied methodology, no leakage management is required for this project activity.

1.19.2 Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

1.19.3 Further Information

No further information is relevant to this project description.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Stakeholder Identification	Stakeholder identification was conducted in 2014 during the Environmental Impact Assessment (EIA) process conducted for the FX Group Lothair facility, in accordance with the National Environmental Act (Act No 107 of 1998) and NEMA regulations 2006, as amended. A public meeting announcement for 18 March at Silindile Community hall was made through various communication routes. An invitation for registration of Interested and Affected parties was announced, which allowed for the identification of stakeholders likely impacted by the process. The registered stakeholders consisted of: - Residents of Sindilile, Lothair - Members of the Lothair business community - FX Group board and staff - Local and environmental authorities
Legal or customary tenure/access rights	The project is located on privately owned land, where the FX group Lothair wood panel plant and biomass to power plant are built on portion 60 of the farm Lothair 124 IT, Lothair, Mpumalanga province. There are no conflicting rights to access the territories, with no land-claims lodged against the territory, and no conflicting rights to access resources required by the project activity.
Stakeholder diversity and changes over time	The stakeholders involved in or impacted by the project consist of four main categories: local communities, local businesses authorities and project proponents. Local communities primarily consists of the nearby township Sindilile. Socially and economically the project activity diversifies jobs and skills within the local communities, by creating new jobs for residents of Sindilile. Local businesses, primarily nearby sawmills, are related to the project activity through their provision of biomass waste residues which are consumed by the project. They benefit from the project by attaining an income stream for their waste residues.
Expected changes in well-being	In the absence of the project activity, the waste biomass residues would be dumped in landfills and left to decay. The well-being of local communities is expected to improve due to the provision of jobs and income streams from biomass residues transportation and utilization, as compared to the baseline scenario. The environmental impacts on stakeholder well-being are minimal, with all impacts during construction and operation categorized as 'low' or 'medium' according to the environmental impact assessment conducted for the site.
Location of stakeholders	The stakeholders included primarily residents of Silindile, as well as members of the Lothair business community, represented by nearby sawmills. Most affected stakeholders are located within the village of Lothair, Mpumalanga.
Location of resources	The primary resource consumed by the project is waste biomass residues which is sourced from local sawmills in Lothair, Mpumalanga and the surrounding area, as well as other FX Group facilities.

2.1.2 Stakeholder Consultation and Ongoing Communication

Date of stakeholder consultation	18-March-2014
Stakeholder engagement process	Public meeting announcements were made through the publication of notice in the Highveld tribune newspaper on 4 March 2014. Additionally, notice posters were placed at the site of the project, in Lothair CBD and at various points in Silindile township, including the local municipality offices. A4 size notice leaflets were placed at various businesses, the police station and other public places. A number of A5 pamphlets were also provided for distribution at various points. This process was conducted with active participation and assistance by local ward councillors following an informative meeting on 12 February 2014 which included members of local ward committees. The notices announced the public meeting and invited interested and affected parties to register. The meeting was organized with the assistance of the ward 15 councillor, Mr S.L Jele and his colleagues. The public meeting was conducted on 18 March, 2014 at Silindile community hall. The following information posters were placed on the wall of the hall for the general information of the attendees of the public meeting: • Explanation of the project process with details of EIA procedures • Explanations of the social and economic impacts of the project • Explanations of the considered environmental impacts of the project • The proposed plant layout and site map In addition to these posters, an agenda for the meeting and description of the planned project were distributed to attendees. The main presentation was presented in English, an interpreter who could answer and respond to questions in isiZulu was present. The attendance at the meeting consisted primarily of residents of Silindile as well as some members of the Lothair business community. Total attendance at the meeting was estimated at 400-450 people. Of these, more than 300 were registered while the remainder joined after registration closed. During the meeting, it was emphasized that the purpose of the meeting was to inform the affected people of the positive and negative potential impacts of the project and give them an opportunity to voice their feedback, questions, queries, suggestions and objections. The primary subject of the meeting was socio-economic and environmental impacts. All questions, concerns and feedback, including those not related to the project but voiced at the meeting, were recorded alongside the response from the speaker. The full results of the local stakeholder consultation have been captured in a public participation process report, made available to the auditor.
Consultation outcome	The key discussion points which emerged from the local stakeholder consultation included:

	• The overwhelming interest of local residents in the possibility of directly and indirectly created jobs was expressed; • Expectation was expressed by the local community that job opportunities would result in the development of local skills; • Concern was expressed about the possible negative effects of the project on the already limited electricity supply in the area. These concerns were addressed by informing residents that the plant would generate its own electricity and would not affect Eskom electricity supply; • Disappointment was voiced that the implementation of the project, and the associated job creation, would only occur after a year, provided that the project was approved. No written responses or objections were received from any party following the public meeting. No objections were received in any capacity. The full results of the local stakeholder consultation have been captured in a public participation process report, made available to the auditor.
Ongoing communication	A complaints register has been kept at the gate of the facility, enabling any affected parties to lodge complaints or concerns. The register is audited by a third party as part of the environmental compliance audit reports, conducted quarterly by an accredited environmental assessment practitioner. To date, no complaints have been received. Additionally, FX group holds regular meetings with local community bodies as well as trade unions, providing a mechanism for stakeholder feedback and ongoing communication.
Stakeholder input	No written responses or objections were received from any party following the public meeting. No objections were received in any capacity. The full results of the local stakeholder consultation have been captured in a public participation process report, made available to the auditor. The local stakeholders expressed great interest in the jobs made available by the project, due account was taken of this input and incorporated into the hiring process, in which local community members were prioritised.

2.1.3 Free Prior and Informed Consent

Obtaining consent	The project activity is undertaken on privately owned land, purchased through a formal land purchase process, in which no land claims or conflicts occurred. Consequently, no IPs, LCs or customary rights holders were affected or required consent. Local stakeholder consultation was conducted and a platform was given to interested and affected parties to voice concerns or objections. Additionally, a complaints register has been actively maintained and audited at the project site.
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Outcome of FPIC

Throughout the process of local stakeholder consultations, no objections were received and no complaints have been lodged in the maintained complaints register throughout the lifetime of the project.

2.1.4 Grievance Redress Procedure

Development process

As required by the Atmospheric emissions license (reference number: Msukaligwa/The FX Group (Pty) Ltd/0043/2023/F01) held by the FX Group Lothair facility, a Complaints register is maintained at the site. The complaint's register is made available for inspection by environmental authorities and contains the name, address and telephone number, date and time when the complaint was registered. The register further provides space for noise, dust and offensive odor complaints.

Grievance redress procedure

The number and details of complaints logged are reported to the licensing authority on a monthly basis.

2.1.5 Public Comments

Comments received

Has not started yet.

Actions taken

Comments and responses will be added in the next version of the PD.

2.2 Risks to Stakeholders and the Environment

2.2.1 Management Experience

The staff at FX Group Lothair facility has extensive experience in the management and operation of power plants, personnel management and human resource management. The Senior Process Controller has experience in operating and monitoring electricity generating plants and co-ordinating and communicating with co-workers to operate parts of the plant. The Lothair Site Manager has experience with plant management and operations management. The Production manager has experience in Process Control, shift management and quality management. The Human Capital Officer has experience in maintaining relationships between employers and employees and trade unions, which include wider communities.

2.2.2 Risk Assessment

	Risks identified	Mitigation or preventative measure(s) taken
Natural and human-induced risks to	No risk identified	The site operations are compliant with the Operational Health and

stakeholders' wellbeing		Safety Act (OSHA) (Act no. 85 of 1993). The project activity was assessed for environmental impacts through an EIA assessment.
Risks to stakeholder participation	No risk identified	Prior to construction, a public meeting process was undertaken to allow for stakeholder feedback. Since construction, a complaints register has been actively maintained and regular meetings with local community groups and trade unions have been conducted.
Working conditions	Risks due to working conditions could include bodily injury due to operation of heavy machinery and health risks due to dust, smoke, ash or other air pollutants during construction or otherwise. Fire hazards represent another risk to working conditions.	Health and Safety management is applied on site during operation. All workers are provided with the necessary PPE and standard safety practices are applied. The site operations are compliant with the Operational Health and Safety Act (OSHA) (Act no. 85 of 1993).
Safety of women and girls	No risk identified	The proponent is governed by and abides by all national statutes and regulations which protect the safety and well-being of women and girls. No complaints have been received relating to the mistreatment of women and girls.
Safety of minority and marginalized groups, including children	No risk identified	The proponent is governed by and abides by all national statutes and regulations which protect the safety and well-being of children and marginalized groups. No complaints have been received relating to the mistreatment of children.
Pollutants (air, noise, discharges to water, generation of waste, and release of hazardous materials and chemical pesticides and fertilizers)	Noise pollution is identified as medium risk in both mitigated and unmitigated scenarios during the construction phase. Stormwater contamination during construction is identified as a medium risk without mitigation and low risk with mitigation. During the construction phase, contamination with fuels and oils is identified as low risk with mitigation and medium risk without mitigation.	Appropriate staff training and ensuring proper procedures are identified as mitigation measures. Strict rules regarding potential contaminants are identified as a prevention measure. Designation of demarcated areas where fuels and oil are to be handled and preparing the ground so that spillage is contained are identified as prevention measures.

	Contamination with cement and other building materials is identified as medium risk without mitigation and low risk with mitigation.	Areas where cement and building materials are spilled must be regularly cleaned and any contaminated soil removed.
	During the operational phase, leakage of sanitary water and sewage leakage is identified as low risk.	Maintenance and regular cleaning of the sanitation water and sewage treating facilities will be used as mitigation measures.
	During operation, contamination of water with chemicals is identified as medium risk under no mitigation and low risk with mitigation.	Mitigation measures include: designing delivery and transfer areas to contain leakages and having sediment pits for contaminated water.
	During the operation phase, contaminated solid waste is identified as medium risk with no mitigation and low risk with mitigation	Having a contingency plan for dealing with solid waste management and keeping stockpiles of absorbent and neutralizing materials.
	During the operational phase, the risk of effect on air quality is identified as medium under no mitigation and low under mitigation.	The use of resin with safe emission characteristics will be used to mitigate the risk.
	Noise inside of production buildings and in open spaces is identified as medium risk with no mitigation and low risk with mitigation.	Measures to minimize noise from machinery and designing buildings to minimize noise leakage will be implemented with precautions to minimize noise of outdoor equipment.
	Vehicle noise is identified as high risk with no mitigation and medium risk with mitigation	To mitigate this risk, only vehicles in good condition will be permitted onto the premises.

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

	Risks identified ²	Mitigation or preventative measure(s) taken
Discrimination	No risk identified	No issues of discrimination have been reported during the project activity.
Sexual harassment	No risk identified	No reports of sexual harassment have been filed during the project activity.
Equal pay for equal work	No risk identified	The proponent compensates their staff based on experience and qualifications, irrespective of gender or race.
Gender equity in labor and work	No risk identified	The proponent compensates and hires staff based on experience and skills, irrespective of gender.
Forced labor	No risk identified	All employees of FX group are employed under a contractual basis.
Child labor	No risk identified	The proponent acts in line with national labor regulations, including the Basic conditions of employment act (No. 75 of 1997) as amended (No. 11 of 2002). Prior to employment, the proponent verifies the age of all hired individuals according to their identity documents, ensuring they are 18 years or older.
Human trafficking	No risk identified	No human trafficking is conducted by the proponent. All employees are hired on a consensual, contractual basis.

2.3.2 Human Rights

Risks identified	Mitigation or preventative measure(s) taken
No risk identified	The project activity is conducted on land which was privately owned prior to and during the implementation of the project activity. Therefore, no IPs, LCs or customary rights holders are impacted by the project activity. The project is governed and abides by South African Human rights laws and regulations.

² The identified risks and commensurate mitigation or preventative measure(s) for forced labor, child labor, and human trafficking, must be inclusive of staff and contracted workers employed by third parties.

2.3.3 Indigenous Peoples and Cultural Heritage

Risks identified	Mitigation(s) or preventative measure taken
No risk identified	A heritage impact assessment was carried out for the entire portion 60 of the farm Lothair 124 IT, on which the project site is located. This assessment was conducted in accordance with the National Heritage and Resources Act (Act 25 of 1999). Although some objects and features of sensitive nature were found on the overall farm area, none of them are near the project site. It was therefore concluded that regarding Act 25 of 1999, no objects, structures or graves of potential relevance are close enough to the project site. Therefore there is no risk to cultural heritage posed by the project activity. The project activity is conducted on land which was privately owned prior to and during the implementation of the project activity. Therefore, no risk is associated with the rights of indigenous peoples or customary rights holders.

2.3.4 Property Rights

Risks identified	Mitigation or preventative measure(s) taken
No risk identified	During purchase of land in South Africa, the state has oversight over the process and the Surveyor general cross-references the land against any existing land claims. No land claims have been registered against the land and ownership is defined by contractual private land purchase. Therefore no property rights are inflicted upon by the project activity.

2.3.5 Benefit Sharing

Process used to design the benefit sharing plan	No property rights are impacted by the project activity, as per section 2.4.4, therefore this section is not applicable.
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Summary of the benefit sharing plan	No property rights are impacted by the project activity, as per section 2.4.4, therefore this section is not applicable.
Approval and dissemination of benefit sharing plan	No property rights are impacted by the project activity, as per section 2.4.4, therefore this section is not applicable.

2.4 Ecosystem Health

	Risks identified	Mitigation or preventative measure(s) taken
Impacts on biodiversity and ecosystems	In the construction phase, disturbance of surrounding areas is identified as low risk in both mitigated and unmitigated scenarios. During the operational phase, Fires outside the plant perimeter is identified as low risk	Prior assessment of foreseeable problems and identification of prevention measures have been stated as mitigation measures. Keeping the perimeter free of combustible material and having equipment and trained personnel for veld fires will be used as an mitigation measure.
Soil degradation and soil erosion	Danger of soil erosion has been identified as a medium risk during the construction phase under unmitigated scenarios and low risk under mitigated scenarios.	The regular inspection of areas where storm water run-off occurs and maintaining these areas in a good condition.
Water consumption and stress	During the construction and operational phase, water wastage is identified as a low risk with and without mitigation.	The mitigation measure implemented is ensuring a good water distribution system that supplies areas that need the resource the most. There is also a responsible person to look after water-related matters. During the operational phase, strict control over water usage will be implemented.

2.4.1 Rare, Threatened, and Endangered Species

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☐ Yes

☒ No

	Risks identified	Mitigation or preventative measure(s) taken
Habitats for rare, threatened, and endangered species	No risk identified	The flora and fauna assessment conducted as part of the EIA process indicated that no sensitive fauna or flora species were documented and no important bird species areas are found within the Lothair farm vicinity.
Areas for habitat connectivity	No risk identified	While wetland areas exist on the Lothair farm, they are sufficiently far from the project activity (1 km) area, to not be affected, as per the environmental impact assessment conducted for the project.

2.4.2 Introduction of Species

This section is not applicable as no new species are introduced in the project activities.

2.4.3 Ecosystem Conversion

This is not applicable as the project is not an AFOLU project.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	CDM: AMS-I.F	AMS-I.F: Renewable electricity for captive use and mini-grid	5.0
Tool	CDM: TOOL03	TOOL03: Tool to Calculate Project or Leakage CO2 Emissions from Fossil Fuel Combustion	03.0
Tool	CDM: TOOL12	TOOL12: Project and Leakage Emissions from Transportation of Freight	01.1.0

Tool	CDM: TOOL16	TOOL16: Project and leakage emissions from biomass	5.0
Tool	CDM: TOOL21	TOOL21: Demonstration of additionality of small-scale project activities	13.1
Tool	CDM: TOOL23	Additionality of first-of-its-kind project activities	3.0
Methodology	CDM: AMS-I.D	AMS-I.D: Grid connected renewable electricity generation	18.0

3.2 Applicability of Methodology

Methodology ID	Applicability condition	Justification of compliance
AMS-I.F- Version 05.0- Renewable electricity generation for captive use and mini-grid	AMS-I.F- Paragraph 3: This methodology is applicable for project activities that (a) Install a new power plant at a site where there was no renewable energy power plant operating prior to the implementation of the project activity (Greenfield plant	The project activity revolves around the installation of a new, greenfield power plant at a site where there was no renewable energy power plant operating prior to the implementation of the project activity, thereby meeting the applicability of option (a)
AMS-I.F- Version 05.0- Renewable electricity generation for captive use and mini-grid	AMS-I.F- Paragraph 4: Table 2 Project displaces grid electricity consumption (e.g. grid import) and/or captive fossil fuel electricity generation at the user end (excess electricity may be supplied to a grid)	The project activity displaces grid electricity. In the absence of the project activity, the wood panel plant would utilised electricity from the South African national grid
AMS-I.F- Version 05.0- Renewable electricity generation for captive use and mini-grid	AMS-I.F- Paragraph 8 Combined heat and power (co-generation) systems are not eligible under this category.	The project activity revolves around a electrical power-only plant and does not implement co-generation of heat and power.
AMS-I.F- Version 05.0- Renewable electricity generation for captive use and mini-grid	AMS-I.F- Paragraph 11 In the case the project activities utilizes biomass, the “TOOL16: Project and leakage emissions from biomass” shall be applied to determine the relevant project emissions from the cultivation of	The project activity utilizes biomass residues, therefore TOOL16 shall be applied to determine the relevant project emissions from the cultivation of biomass and the utilization of biomass residues.

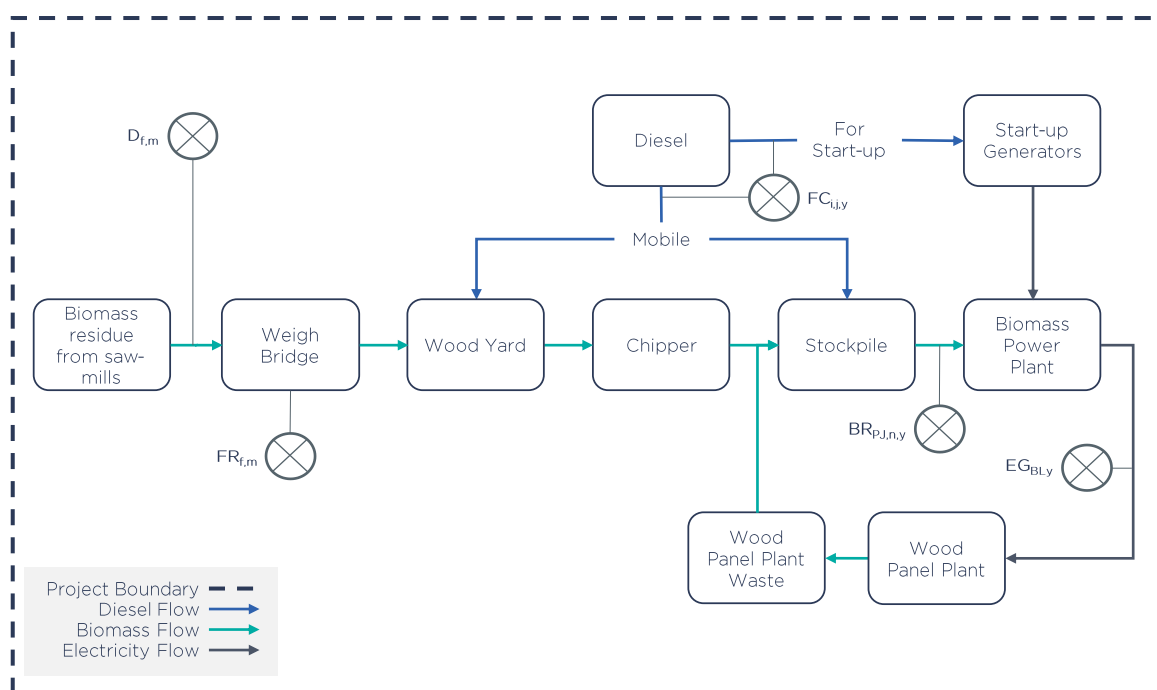
	biomass and the utilization of biomass or biomass residues.	
TOOL16 – Version 05.0- Project and leakage emissions from biomass	The tool is also applicable if biomass residues are consumed in a CDM project activity, and the biomass residues can be utilized after processing or without processing. These could be: (a) Procured by the project proponents; or (b) The result of an agro-industrial process under the control of the project proponents.	The project activity consumes biomass residues, which are procured by the project proponent as captured in paragraph 8 (a) of the tool and are the result of agro-industrial processes which are under the control of the project proponent, as captured in paragraph 8 (b). For these reasons, the project activity meets the applicability conditions of this tool.
TOOL03- Version 03.0- Tool to Calculate Project or Leakage CO2 Emissions from Fossil Fuel Combustion	This tool provides procedures to calculate project and/or leakage CO2 emissions from the combustion of fossil fuels. It can be used in cases where CO2 emissions from fossil fuel combustion are calculated based on the quantity of fuel combusted and its properties. Methodologies using this tool should specify to which combustion process j this tool is being applied.	The project activity involves the use of small amounts of fossil fuels for the start-up of the biomass boiler as well as the use of mobile equipment for the handling of biomass on site. For this reason, this tool is applicable.
TOOL12 Version 01.1.0 – Project and Leakage Emissions from Transportation of Freight	This tool is applicable to project activities which involve freight transportation by road and where transportation is not the main project activity. This tool is not applicable to project activities where transportation is the main source of greenhouse gases emissions. This tool does not provide procedures to estimate baseline emissions from road transportation of freight. The tool only provides to determine CO2 emissions.	The Project will involve the transportation of biomass residues from sites beyond the project boundary to the project site using heavy vehicle trucks. However, transportation is not the main source of greenhouse gas emissions. Therefore, this tool is applicable. The Project will not rely on railway transport for the transportation of biomass to the plant.
TOOL21 Version 13.1- Demonstration of additionality of small-scale project activities	This methodological tool provides a general framework for demonstrating and assessing additionality and is applicable to a wide range of project types.	This methodological tool is applicable to a wide range of project types. Given the project activity is small-scale, this tool is applicable.
TOOL23 Version 03.0- Additionality of	This methodological tool is applicable to project activities that wish to use the "first-of-itskind"	AMS-I.F and its accompanying tools allow the use of the "first-of-

first-of-its-kind project activities	Approach to demonstrate its-kind" approach for additionality	approach for demonstrating additionality.
AMS-I.D Version 18.0- Grid connected renewable electricity generation	This methodology is applicable to project activities that: (a) Install a Greenfield plant;	In accordance with AMS-I.F the procedures provided in AMS-I.D is applied for the calculation of the emission factor of an electric grid for the determination of baseline emissions.

3.3 Project Boundary

As per paragraph 18 of Methodology AMS-I.F version 05.0, the spatial extent of the project boundary includes industrial, commercial facilities consuming energy generated by the system.

Hence, the project boundary of the FX Group biomass to energy project includes the FX Group Lothair wood panel plant, the biomass processing facilities (Chipper), the stockpile where biomass residues will be temporarily stored, the biomass power plant, the transportation system used to move biomass to the project site and the start-up generators. The project boundary, alongside the project equipment, systems and flows of mass and energy are depicted in the figure below.



The relevant GHG sources, sinks and reservoirs for the project and baseline scenarios are presented in the table below.

Source		Gas	Included?	Justification/Explanation
Baseline	Electricity generation	CO ₂	Yes	Main emission source.
		CH ₄	No	Excluded for simplification. This is conservative.
		N ₂ O	No	Excluded for simplification. This is conservative.
		Other	No	Excluded for simplification.
	Aerobic or anaerobic decay of surplus biomass residues	CO ₂	No	It is assumed that CO ₂ emissions from surplus biomass residues do not lead to changes in the carbon pools in the LULUCF sector.
		CH ₄	No	Excluded for simplification. This is conservative.
		N ₂ O	No	Excluded for simplification. This is conservative.
		Other	No	Excluded for simplification.
Project	On-site fossil fuel consumption	CO ₂	Yes	May be an important emission source.
		CH ₄	No	Excluded for simplification. This emission source is assumed to be very small.
		N ₂ O	No	Excluded for simplification. This emission source is assumed to be very small.
		Other	No	Excluded for simplification.
	On-site and off-site transportation of biomass residues.	CO ₂	Yes	May be an important emission source.
		CH ₄	No	Excluded for simplification. This emission source is assumed to be very small.
		N ₂ O	No	Excluded for simplification. This emission source is assumed to be very small.
		Other	No	Excluded for simplification
	Cultivation of land to produce biomass residue feedstock	CO ₂	No	Biomass used in the project activity is not sourced from a dedicated plantation.
		CH ₄	No	Biomass used in the project activity is not sourced from a dedicated plantation.
		N ₂ O	No	Biomass used in the project activity is not sourced from a dedicated plantation.
		Other	No	Biomass used in the project activity is not sourced from a dedicated plantation.
	Combustion of biomass for electricity	CO ₂	No	It is assumed that CO ₂ emissions from surplus biomass residues do not lead to changes in the carbon pools in the LULUCF sector.
		CH ₄	No	This emission source is excluded because CH ₄ emissions from uncontrolled burning or

Source	Gas	Included?	Justification/Explanation
			decay of biomass residues in the baseline scenario are excluded
	N ₂ O	No	Excluded for simplification. This emission source is assumed to be very small
	Other	No	Excluded for simplification.

3.4 Baseline Scenario

Without the implementation of the 4.5MW biomass plant, the FX Group would have to rely on a grid electricity line supplied by Eskom. However, the existing line is insufficient to meet the required energy output, and upgrading to a larger supply line would incur additional environmental impacts. Additionally, Eskom generated electricity is highly carbon-intensive, dependant on coal-fired power stations. Hence, the baseline scenario would generate more carbon emissions, due to the dependence on carbon-intensive electricity from the national utility Eskom. The alternative baseline scenario for each relevant biomass category is presented in Table format in section 4.3.

3.5 Additionality

The AMS-I.F methodology does not prescribe a specific tool to demonstrate additionality. The project proponent has elected to apply the CDM TOOL23: “*Additionality of first-of-its-kind project activities*” and CDM TOOL21: “*Demonstration of additionality of small-scale project activities*”.

The Project meets the first-of-its-kind” assessment criteria outlined in TOOL23 and is therefore additional. According to the Tool, a proposed project activity is the first-of-its-kind if:

- The project is the first in the applicable geographical area that applies a technology that is different from technologies that are implemented by any other project, which are able to deliver the same output and have started commercial operation in the applicable geographical area before the project design document is published for global stakeholder consultation or before the start date of the proposed project activity, whichever is earlier;
- The project implements one or more of the measures;
- The project participants selected a crediting period for the project activity that is “a maximum of 10 years with no option of renewal”.

The proposed project is located within the Republic of South African (the applicable geographical area) and consists of a change of energy source from grid electricity to sustainable biomass residue (the applied measure) with a capacity of 4.5M. According to TOOL 23’s definition of different technologies, the project is the first of its kind in the applicable geographical area, as elaborated by comparison to existing projects in the applicable geographical area in section 3.5.2.

Therefore the project is the first-of-its-kind micro-scale biomass residue to energy project in South Africa. The project proponent has selected a crediting period for the project activity that is “a maximum of 10 years with no option of renewal”. Herewith the Project activity meets all the additionality criteria as set out in TOOL23.

On top of being additional from a first-of-its-kind perspective, the project activity faced substantial barriers to implementation, to the extent that the project would not have materialised in the absence of carbon finance making it additional from a barrier perspective. As this project activity is micro-scale (below 5MW). TOOL21 version 13.1 requires an explanation to show that the project activity would not have occurred anyway due to at least one of the following barriers:

- (a) Investment barrier
- (b) Technological barrier
- (c) Barrier due to prevailing practice
- (d) Other barriers: such as institutional barriers or limited information, managerial resources, organizational capacity, financial resources, or capacity to absorb new technologies.

The project proponent identifies and analyses “Technological barrier” and “Barrier due to prevailing practice” as the main challenges faced by the project, which would have prevented the project from materialising in the absence of a carbon finance market. The analysis of each of these barriers, with supporting information from the applicable geographical area are presented in section 3.5.2.

3.5.1 Regulatory Surplus

Is the project located in an UNFCCC Annex 1 or Non-Annex 1 country?

- ☐ Annex 1 country ☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

- ☐ Yes ☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

- ☐ Yes ☐ No

The project is located in a Non-Annex 1 country and the project activities are not mandated by a law, statute, or other regulatory framework. Question 3 is therefore not applicable.

3.5.2 Additionality Methods

The AMS-I.F methodology does not prescribe a specific tool to demonstrate additionality. The project proponent has elected to apply the CDM TOOL23: “*Additionality of first-of-its-kind project activities*” and CDM TOOL21: “*Demonstration of additionality of small-scale project activities*”.

The Project meets the first-of-its-kind” assessment criteria outlined in TOOL23 and is therefore additional. According to the Tool, a proposed project activity is the first-of-its-kind if:

- a) The project is the first in the applicable geographical area that applies a technology that is different from technologies that are implemented by any other project, which are able to deliver the same output and have started commercial operation in the applicable geographical area before the project design document is published for global stakeholder consultation or before the start date of the proposed project activity, whichever is earlier;
- b) The project implements one or more of the measures;
- c) The project participants selected a crediting period for the project activity that is “a maximum of 10 years with no option of renewal”.

The proposed project is located within the Republic of South African (the applicable geographical area) and consists of a change of energy source from grid electricity to sustainable biomass residue (the applied measure) with a capacity of 4.5M. Tool 23 defines different technologies as technologies that deliver the same output and differ by at least one of the following (as appropriate in the context of the measure applied in the proposed project activity and applicable geographical area):

- a) Energy source/fuel (example: energy generation by different energy sources such as wind and hydro and different types of fuels such as biomass and natural gas);
- b) Feed stock (example: production of fuel ethanol from different feed stocks such as sugar cane and starch, production of cement with varying percentage of alternative fuels or less carbon-intensive fuels);
- c) Size of installation (power capacity)/energy savings:
 - I. Micro (as defined in paragraph 24 of decision 2/CMP.5 and paragraph 39 of decision 3/CMP.6);
 - II. Small (as defined in paragraph 28 of decision 1/CMP.2);
 - III. Large.

The table below identifies biomass-to-energy projects in South Africa and provides an analysis based on the above technology criteria. It shows that there are currently no similar technologies applied within the country’s geographic boundaries. For this reason, the project is considered to be first-of-its-kind in the country.

Project name	Technology	Sources	Status	Comparable?
Mkuze Biomass Project	16.5MW Power-only production for supply to the grid	IPP office	permitting stage	No (large-scale)
Mondi Richards Bay Biomass Project	Fuel switch for heat production for own use	CDM	Active	No (heat production for captive usage)
Lomati Biomass Power Generation Project	9MW Power-only production for supply to the grid	CDM	Abandoned	No (small-scale)
Tugela Mill Fuel Switching Project	Fuel switch for heat production for own use	CDM / VCS	Abandoned	No (small-scale / heat production of captive usage)
Howick Wood Pallet Plant	5MW Power-only production for own use and supply to the grid	iiied	Abandoned prior to full commissioning	No (small-scale)
Tstsikamma Biomass Plant	6MW Power-only production for own use and supply to the nearby communities and the grid	iiied	closed down	No (small-scale)
Grahamstown Invasive Biomass Power Project	3.5MW Power-only production for supply to the grid	CDM	Abandoned prior to commissioning	No (has not started commercial operation)
Ngodwana Biomass to Energy plant	25MW Power-only production for supply to the grid	VCS	operational	No (large-scale)

The table above indicates that the project is the first-of-its-kind micro-scale biomass residue to energy project in South Africa. The project proponent has selected a crediting period for the project activity that is “a maximum of 10 years with no option of renewal”. Herewith the Project activity meets all the additionality criteria as set out in TOOL23.

On top of being additional from a first-of-its-kind perspective, the project activity faced substantial barriers to implementation, to the extent that the project would not have materialised in the absence of carbon finance making it additional from a barrier perspective. As this project activity is micro-scale (below 5MW). TOOL21 version 13.1 requires an explanation to show that the project activity would not have occurred anyway due to at least one of the following barriers:

- (a) Investment barrier
- (b) Technological barrier
- (c) Barrier due to prevailing practice
- (d) Other barriers: such as institutional barriers or limited information, managerial resources, organizational capacity, financial resources, or capacity to absorb new technologies.

Additionality for this project activity will be demonstrated due to the technological barrier and the barrier due to prevailing practise.

Technological barrier

In South Africa, previous attempts to implement biomass energy projects have faced numerous challenges. Several similar sized biomass initiatives, including the Howick Wood Pallet Plant, the Tstsikamma Biomass Plant, and the Grahamstown Biomass Plant, have encountered significant difficulties and ultimately failed. These projects demonstrate that biomass to energy projects, although technically feasible in the country, have experienced many more challenges in implementation.

1 Howick Wood Pallet Plant – 5 MW Plant³

The Howick Wood Pallet Plant produced biomass primarily for export to the European wood energy market. Within the initial project design there was also a 5MW biomass electricity plant, both to supply the pallet processing plant, and to sell surplus energy to neighboring industries and Eskom. The power plant was never fully commissioned because the project developer could not secure a favorable power purchase agreement from Eskom for the surplus electricity. The pallet processing plant had to be expanded its feedstock supply radius to 97km to meet the required volume. This adjustment led to increased costs associated with road haulage, cleaning, and drying of the raw material. Ultimately, the project's high maintenance (as the available fuel could not be adequately processed by the plant) and logistical costs led to its liquidation.

2 Tstsikamma Biomass Plant – 6MW Plant ⁴

The Tstsikamma Biomass Plant utilized waste from nearby sawmills. It was intended to power the sawmill and neighboring communities, with potential to supply the national grid. However, Eskom, at that time, believed it had sufficient capacity and declined to pay for the additional connection. In 2012, after a dispute settlement, the plant was taken over by a new company that did not incorporate the sawmill into its operations. Consequently, the sawmill closed, and the biomass plant was also shut down. Moreover, the plant required costly upgrades that were deemed uneconomical.

³ South African biomass energy: little heeded but much needed

<https://www.iied.org/17165iied#:~:text=South%20Africa%20badly%20needs%20more,agreed%20Long%20Term%20Mitigation%20Scenarios.>

⁴ South African biomass energy: little heeded but much needed

<https://www.iied.org/17165iied#:~:text=South%20Africa%20badly%20needs%20more,agreed%20Long%20Term%20Mitigation%20Scenarios.>

3 Grahamstown Biomass Plant – 3.5MW Plant ⁵

The Grahamstown Biomass Plant was successfully registered as a CDM project but never commissioned. The project activity demonstrated additionality through the financial barrier, however, the opportunity for carbon credit revenue was never realized. Although several attempts were made to reach-out to the project developer and determine which barrier led to the demise of the project, no contact was established.

4 Comparison with Other Projects

It is also noteworthy that other biomass projects in South Africa are significantly larger than the Project Activity. A comparative table outlining the sizes of these projects further underscores the difference in scale.

Name	Size	Source
Mondi Richards Bay	75 MW	CDM ⁶
Lomati	10.1 MW	CDM ⁷
Mkuze	16.5 MW	REIPPPP ⁸
Ngodwana	25MW	VCS ⁹

Barrier due to prevailing practise

Although there are biomass residues from board manufacturing, renewable biomass to energy projects are not common in the industry. Prevailing practise is to use grid electricity or energy from the combustion of fossil fuels.

For example, one of the largest suppliers of particleboard and MDF nationally is PG Bison, which does not include renewable electricity in its environmental policy nor its “Grow with Us. Our responsible and sustainable practices.” sustainability reporting. PG Bison recently completed a new MDF production line at their Mkhondo manufacturing facility. The announcement of the investment¹⁰, made in November 2020, included R1.42 Billion in investment for the latest

⁵ Project 8372 : Grahamstown Invasive Biomass Power Project
https://cdm.unfccc.int/Projects/DB/KBS_Cert1353665832.64/view

⁶ Project 0966 : Mondi Richards Bay Biomass Project
<https://cdm.unfccc.int/Projects/DB/SGS-UKL1172139189.07>

⁷ Project: 7476 Lomati Biomass Power Generation Project in Mpumalanga Province – History
<https://cdm.unfccc.int/Projects/DB/DNV-CUK1348731809.14/history>

⁸ RE_BM_0149_019 Mkuze
<https://www.ipp-projects.co.za/ProjectDatabase>

⁹ Ngodwana Biomass to Energy Project (1982)
<https://registry.terra.org/app/search/VCS>

¹⁰ PG BISON ANNOUNCES ADDITIONAL INVESTMENT IN SUPPORT OF INDUSTRY GROWTH.
https://pgbison.co.za/sites/default/files/2020-11/PG_Bison_Investment_Announcement_17_Nov_2020.pdf

technology, with no mention of renewable energy. Further, the May 2024 press release confirming the commissioning of the plant, confirms a productive partnership with Eskom (the national grid operator)¹¹, indicating use of carbon intensive grid electricity for the new plant.

The above confirms that there are material technological barriers and barriers due to prevailing practise making the project additional in that it would not be able to overcome these barriers in the absence of revenue potential from the sale of carbon credits.

3.6 Methodology Deviations

No methodology deviations are applied to this project.

¹¹ First Board Produced at PG Bison's New Mkhondo MDF Line. <https://pgbison.co.za/blog/first-board-produced-at-pg-bisons-new-mkhondo-mdf-line>

4 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

The baseline emissions are calculated as the product of the amount of electricity displaced with the electricity produced by the renewable generating unit and an emission factor. The formula for determining baseline emissions is presented below:

$$BE_y = EG_{BL,y} \times EF_{CO_2,y} \quad (\text{Equation 1})$$

Where:

Variable		Definition	UoM
BE_y	=	Baseline emissions in year y	tCO ₂
$EG_{BL,y}$	=	Quantity of net electricity displaced as a result of the implementation of the project activity in year y	MWh
$EF_{CO_2,y}$	=	Grid emission factor, calculated as per the procedures provided in AMS-I.D	tCO ₂ /MWh

Baseline emissions are determined through the following steps:

Step 1: Determination of $EF_{CO_2,y}$

The emission factor of an electric grid shall be calculated according to the procedures provided in AMS-I.D version 18.0 “Grid connected renewable electricity generation”. Accordingly, the emission factor is calculated in a transparent and conservative manner by calculating the weighted average emissions (in t CO₂/MWh) of the current generation mix. The data of the year in which project generation occurs must be used (option b).

Calculations shall be based on data from an official source (where available) and made publicly available. This project elects to use option (b), determining the weighted average emissions of the current generation mix, based on publicly available data from the official national utility-Eskom.

Step 2: Determine $EG_{BL,y}$

The quantity of net electricity displaced as a result of the implementation of the project activity in the year y is determined by measuring the net-generation of the biomass plant in year y.

4.2 Project Emissions

According to AMS-I.F, the CO₂ emissions from on-site consumption of fossil fuels as well as emission sources from projects that utilise biomass and/or biomass residues should be accounted for. Therefore, project emissions are determined through the following steps:

Step 1: Determine project emission sources ($PE_{BRT,y}$, PE_{BRP})

For projects that utilise biomass and/or biomass residues, TOOL16 shall be applied to determine project emission sources. Project emissions are calculated as follows:

$$PE_y = PE_{BC,y} + PE_{BT,y} + PE_{BP,y} + PE_{BRT,y} + PE_{BRP,y} \quad (\text{Equation 2})$$

Where:

Variable		Definition	UoM
PE_y	=	Project emissions during year y	tCO ₂
$PE_{BC,y}$	=	Project emissions resulting from the cultivation of biomass in a dedicated plantation of the project activity that uses biomass	tCO ₂
$PE_{BT,y}$	=	Project emissions resulting from the transportation of biomass	tCO ₂
$PE_{BP,y}$	=	Project emissions resulting from the processing of biomass	tCO ₂
$PE_{BRT,y}$	=	Project emissions resulting from the transportation of biomass residues	tCO ₂
$PE_{BRP,y}$	=	Project emissions resulting from the processing of biomass residues	tCO ₂

The Project only utilizes biomass residues, therefore the emission sources $PE_{BC,y}$, $PE_{BT,y}$ and $PE_{BP,y}$ are not applicable. Therefore, the relevant project emission sources are $PE_{BRT,y}$ and $PE_{BRP,y}$.

Step 1.1: Determine project emissions from transportation of biomass residues ($PE_{BRT,y}$)

TOOL16 prescribes TOOL12 for the determination of project emissions resulting from the transport of biomass residues, where the parameter $PE_{BRT,y}$ corresponds to $LE_{TR,m}$. If the biomass residues are consumed, without further processing, the route shall include only the transport of biomass residues between the biomass processing facility or the biomass generation site and the biomass utilization facility. In this project activity, the biomass is consumed without further processing, therefore $PE_{BRT,y}$ is calculated as follows:

TOOL12 provides two options to determine $PE_{BRT,y}$, namely: (A) Monitoring fuel consumption and (B) Using Conservative default values. This project elects to use option (B), due to the unavailability of fuel consumption data. Therefore, project emissions are calculated as:

$$PE_{TR,m} = \sum_f D_{f,m} \times FR_{f,m} \times EF_{CO2,f} \times 10^{-6} \quad (\text{Equation 3})$$

Where:

Variable		Definition	UoM
$PE_{TR,m}$	=	Project emissions from transportation of freight monitoring period m	tCO ₂
$D_{f,m}$	=	Return trip distance between the origin and destination of freight transportation activity f in monitoring period m	km
$FR_{f,m}$	=	Total mass of freight transported in freight transportation activity f in monitoring period m	t
$EF_{CO_2,f}$	=	Default CO ₂ emission factor for freight transportation activity f	g CO ₂ /t km
f	=	Freight transport activities conducted in project activity monitoring period m	

According to TOOL16 (paragraph 31), as an alternative to the monitoring of parameters needed to calculate the emissions from transportation, project proponents may apply a default emission factor of 0.0142 tCO₂/tonne of biomass. This default value applies only to micro-scale and small-scale project activities and is applied to this small-scale project.

Step 1.2: Determine project emissions resulting from the processing of biomass emissions ($PE_{BRP,y}$)

According to TOOL16, $PE_{BRP,y}$ is calculated as:

$$PE_{BRP,y} = PE_{BRP,electricity,y} + PE_{BRP,fuel,y} + PE_{BRP,CH_4,y} + PE_{BRP,COMP,y} + PE_{BRP,AD,y} + PE_{BRP,ww,y} + PE_{BP,additives,y} \quad (\text{Equation 4})$$

Where:

Variable		Definition	UoM
$PE_{BRP,y}$	=	Project emissions resulting from the processing of biomass residues	tCO ₂
$PE_{BRP,electricity,y}$	=	Project emissions resulting from the consumption of electricity due to thermo-chemical, biological and mechanical processing of the biomass residues in year y	tCO ₂
$PE_{BRP,fuel,y}$	=	Project emissions resulting from the consumption of fossil fuels due to thermo-chemical, biological and mechanical processing of the biomass residues in year y	tCO ₂
$PE_{BRP,CH_4,y}$	=	Project methane emissions resulting from the decay of biomass residues under anaerobic conditions due	tCO ₂

		to thermo-chemical, biological and mechanical processing in year y	
$PE_{BRP,COMP,y}$	=	Project emissions associated resulting from composting due to thermo-chemical, biological and mechanical processing of the biomass residues in year y	tCO ₂
$PE_{BRP,AD,y}$	=	Project emissions resulting from the anaerobic digester due to thermo-chemical, biological and mechanical processing of the biomass residues in year y	tCO ₂
$PE_{BRP,ww,y}$	=	Project emissions resulting from wastewater treatment due to thermo-chemical, biological and mechanical processing of the biomass residues in year y	tCO ₂
$PE_{BP,additives,y}$	=	Project emissions resulting from the use of additives to process the biomass residues in year y	tCO ₂

The project activity utilises renewable electricity, therefore the emission source $PE_{BRP,electricity,y}$ is not applicable. The project activity does not result in the decay of biomass residue nor does it result in the composting of biomass residues, therefore the emission sources $PE_{BRP,CH_4,y}$ and $PE_{BRP,COMP,y}$ are not applicable. Additionally, the project activity does not include anaerobic digestion, waste water treatment or the utilisation of additives, therefore the emission sources $PE_{BRP,AD,y}$, $PE_{BRP,ww,y}$ and $PE_{BP,additives,y}$ are not applicable. Therefore only project emissions from project emissions from fuel consumed due to processing of biomass residues ($PE_{BRP,fuel,y}$) are calculated.

According to TOOL16, emissions resulting from the fuel consumed due to thermo-chemical, biological and mechanical processing of the biomass residues are determined based on the provisions of TOOL03, where the parameter $PE_{BRP,fuel,y}$ corresponds to $PE_{FC,j,y}$ from the tool and is calculated as follows:

Step 1.3: Determination of project emissions from fuel consumed ($PE_{BRP,fuel,y}$)

$$PE_{FC,j,y} = \sum_i FC_{i,j,y} \times COEF_{i,y} \quad (\text{Equation 5})$$

Where:

Variable		Definition	UoM
$PE_{FC,j,y}$	=	CO ₂ emissions from fossil fuel combustion in process j during the year y	tCO ₂ /yr
$FC_{i,j,y}$	=	Quantity of fuel type i in year y	mass or volume unit/yr
$COEF_{i,y}$	=	CO ₂ emission coefficient of fuel type i in year y	tCO ₂ /mass or volume unit
i	=	Fuel types combusted in process j during the year y	

The CO₂ emission coefficient $COEF_{i,y}$ can be calculated using one of two Options in TOOL03, depending on the availability of data on the fossil fuel type i . In turn, this project utilizes option B) whereby the CO₂ emission coefficient ($COEF_{i,y}$) is calculated based on net calorific value and CO₂ emission factor of the fuel type i as follows:

$$COEF_{i,y} = NCV_{i,y} \times EF_{CO2,i,y} \quad (\text{Equation 6})$$

Where:

Variable		Definition	UoM
$COEF_{i,y}$	=	CO ₂ emission coefficient of fuel type i in year y	tCO ₂ /mass or volume unit
$NCV_{i,y}$	=	Weighted average net calorific of the fuel type i in year y	GJ/mass or volume unit
$EF_{CO2,i,y}$	=	Weighted average CO ₂ emission factor of fuel type i in year y	tCO ₂ /GJ
i	=	Fuel types combusted in process j during year y	

Step 2: Determine emissions from on-site fossil fuel consumption ($PE_{FCi,j,y}$)

$PE_{BRP,fuel,y}$ is calculated according to TOOL03 version 03.0 “Tool to calculate project leakage CO₂ emissions from fossil fuel combustion” in the same procedure presented in [Step 1.3](#) above. Therefore, the diesel used in the start-up generators and for biomass processing are recorded separately and calculated together under the parameter $FC_{i,j,y}$ above.

4.3 Leakage Emissions

According to the methodology AMS-I.F, for project activities utilizing biomass and/or biomass residues, the TOOL16 shall be applied to determine the leakage. The Project does not utilize biomass cultivated in a dedicated plantation therefore leakage due to diversion of biomass residues from other applications is determined as follows:

Step 1: Determination of the alternative scenario of the biomass residues in the absence of the project activity

The alternative scenario for the “use”, in the absence of the project activity, of biomass residues that will be used in the underlying project activity shall include:

- (a) B1: The biomass residues are dumped or left to decay mainly under aerobic conditions. This applies, for example, to dumping and decay of biomass residues on fields;
- (b) B2: The biomass residues are dumped or left to decay under clearly anaerobic conditions. This applies, for example, to landfills which are deeper than five meters. This does not apply to biomass residues that are stock-piled or left to decay on fields;
- (c) B3: The biomass residues are burnt in an uncontrolled manner without utilizing it for energy purposes;
- (d) B4: The biomass residues are used for energy or non-energy applications, or the primary source of the biomass residues and/or their fate cannot be clearly identified.

The biomass residues used in the Project activity are defined in the table below.

Table 1: Biomass residues categories

Biomass residue category (k)	Bio-mass residue type	Biomass residue source	Biomass residue fate in the absence of the project activity	Biomass residue use in the project scenario	Biomass residue quantity (tonnes)
1	Saw Dust	On-site production	The biomass residues are dumped or left to decay mainly under aerobic conditions (B1)	Electricity generation on-site (biomass-only boiler)	-
1	Saw Dust	Off-site from an identified mill	The biomass residues are dumped or left to decay under clearly anaerobic conditions (B2)	Electricity generation on-site (biomass-only boiler)	68 620
1	Saw Dust	Off-site from an identified mill	The biomass residues are dumped or left to decay mainly under aerobic conditions (B1)	Electricity generation on-site (biomass-only boiler)	-
1	Saw Dust	Off-site from an unidentified mill	The biomass residues are used for energy or non-energy applications, or the primary source of the biomass residues and/or their fate cannot be clearly identified (B4)	Electricity generation on-site (biomass-only boiler)	-

2	Bark	On-site production	The biomass residues are dumped or left to decay mainly under aerobic conditions (B1)	Electricity generation on-site (biomass-only boiler)	-
2	Bark	On-site production	The biomass residues are dumped or left to decay under clearly anaerobic conditions (B2)	Electricity generation on-site (biomass-only boiler)	730
3	Pine Logs	On-site production	The biomass residues are dumped or left to decay mainly under aerobic conditions (B1)	Electricity generation on-site (biomass-only boiler)	
3	Pine Logs	On-site production	The biomass residues are dumped or left to decay under clearly anaerobic conditions (B2)	Electricity generation on-site (biomass-only boiler)	1 460
4	Pine Chips	On-site production	The biomass residues are dumped or left to decay mainly under aerobic conditions (B1)	Electricity generation on-site (biomass-only boiler)	
4	Pine Chips	On-site production	The biomass residues are dumped or left to decay under clearly anaerobic conditions (B2)	Electricity generation on-site (biomass-only boiler)	3 285
5	Gum Slats	On-site production	The biomass residues are dumped or left to decay mainly under aerobic conditions (B1)	Electricity generation on-site (biomass-only boiler)	
5	Gum Slats	On-site production	The biomass residues are dumped or left to decay under clearly anaerobic conditions (B2)	Electricity generation on-site (biomass-only boiler)	2 190
6	Pine Slats	On-site production	The biomass residues are dumped or left to decay mainly under aerobic conditions (B1)	Electricity generation on-site (biomass-only boiler)	
6	Pine Slats	On-site production	The biomass residues are dumped or left to decay under clearly anaerobic conditions (B2)	Electricity generation on-site (biomass-only boiler)	2 190

Step 2: Calculation of leakage due to diversion of biomass residues

For all categories of biomass residues whose alternative scenario has been identified as B4, leakage emissions are calculated as follows:

$$LE_{BR,Div,y} = EF_{CO_2,LE} \times \sum_n BR_{PJ,n,y} \times NCV_{n,y} \quad (\text{Equation 6})$$

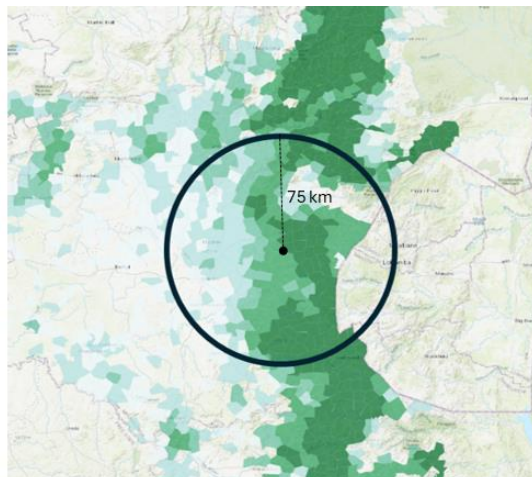
Where:

Variable		Definition	UoM
$LE_{BR,Div,y}$	=	Leakage emissions due to the diversion of biomass residues from other applications in year y	t CO _{2e}
$EF_{CO_2,LE}$	=	CO ₂ emission factor of the most carbon intensive fossil fuel used in the country	t CO ₂ /GJ
$BR_{PJ,n,y}$	=	Quantity of biomass residues of category n used in facilities which are located at the project site and included in the project boundary in year y	tonnes on dry-basis
$NCV_{n,y}$	=	Default CO ₂ emission factor for freight transportation activity f	GJ/tonne of dry matter
n	=	Categories of biomass residues for which B4 has been identified as the alternative scenario	

In addition, TOOL16 requires that the Project Participant demonstrate that there is an abundant surplus of the biomass residue in the project region which is not utilized. For this purpose, demonstrate that the total quantity of that type of biomass residues annually available in the project region is at least 25 percent larger than the quantity of biomass residues which is utilized annually in the project region (e.g. for energy generation or as feedstock), including the project facility;

Biomass residue availability has been mapped comprehensively in the Bioenergy Atlas for South Africa¹² based on so-called mesozones. For the purpose of statistical record keeping relating to social, economic and environmental indicators, South Africa has been divided into 25,000 mesozones. These mesozones are not uniform in shape but aim to be of approximately the same size (i.e. 50 km²). The below figure provides a visual depiction of the available biomass residue by mesozone within a radius 70 km of the project, which is the stated maximum delivery distance for the Project activity.

¹² Bioenergy Atlas for South Africa: <https://catalogue.saeon.ac.za/records/10.15493/BEA.DATA.10000079>



The underlying database with the metadata for availability of biomass residues in tonnes per annum has been made available to the auditor. Based on the analysis of a sample of 10 mesozones in the project region, a surplus of 83 percent above what is being used by the project activity (i.e. $(451\,848 - 78\,475) / 78\,475$) was found.

4.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

Emissions reductions are calculated according to AMS-I.F as follows:

$$ER_y = BE_y - PE_y - LE_y$$

Where:

Variable		Definition	UoM
ER_y	=	Emission reductions in year y	t CO ₂ e/y
BE_y	=	Baseline emissions in year y	t CO ₂ /y
PE_y	=	Project emissions in year y	t CO ₂ /y
LE_y	=	Leakage emissions in year y	t CO ₂ /y

The table below provides the baseline, project, leakage and emission reductions over the full 10 year crediting period.

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated reduction VCU's (tCO ₂ e)	Estimated removal VCU's (tCO ₂ e)	Estimated total VCUs (tCO ₂ e)
12-Oct-2022 to 31-Dec-2022	4 140	299	0	3 842	0	3 842
01-Jan-2023 to 31-Dec-2023	18 585	1 363	0	17 221	0	17 221
01-Jan-2024 to 31-Dec-2024	18 585	1 363	0	17 221	0	17 221
01-Jan-2025 to 31-Dec-2025	18 585	1 363	0	17 221	0	17 221
01-Jan-2026 to 31-Dec-2026	18 585	1 363	0	17 221	0	17 221
01-Jan-2027 to 31-Dec-2027	18 585	1 363	0	17 221	0	17 221
01-Jan-2028 to 31-Dec-2028	18 585	1 363	0	17 221	0	17 221
01-Jan-2029 to 31-Dec-2029	18 585	1 363	0	17 221	0	17 221
01-Jan-2030 to 31-Dec-2030	18 585	1 363	0	17 221	0	17 221
01-Jan-2031 to 31-Dec-2031	18 585	1 363	0	17 221	0	17 221
01-Jan-2032 to 11-Oct-2032	14 409	1 363	0	13 352	0	13 352
Total	185 863	13 629	0	172 233	0	172 233

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	EF _{CO2,y}																																							
Data unit	tCO ₂ /MWh																																							
Description	CO ₂ emission factor for the grid/mini-grid/captive electricity in year y																																							
Source of data	Public data from the national utility (Eskom). The data used in the PD for 2022 and 2023 is based on public available data. Following years are assumed the same as 2023.																																							
Value applied	<table><tr><th colspan="2">Period</th><th>EF_{CO2,y}</th></tr><tr><th>Start</th><th>End</th><th>MWh</th></tr><tr><td>12 10 2022</td><td>31 12 2022</td><td>0.923</td></tr><tr><td>01 01 2023</td><td>31 12 2023</td><td>0.908</td></tr><tr><td>01 01 2024</td><td>31 12 2024</td><td>0.908</td></tr><tr><td>01 01 2025</td><td>31 12 2025</td><td>0.908</td></tr><tr><td>01 01 2026</td><td>31 12 2026</td><td>0.908</td></tr><tr><td>01 01 2027</td><td>31 12 2027</td><td>0.908</td></tr><tr><td>01 01 2028</td><td>31 12 2028</td><td>0.908</td></tr><tr><td>01 01 2029</td><td>31 12 2029</td><td>0.908</td></tr><tr><td>01 01 2030</td><td>31 12 2030</td><td>0.908</td></tr><tr><td>01 01 2031</td><td>31 12 2031</td><td>0.908</td></tr><tr><td>01 01 2032</td><td>11 10 2032</td><td>0.908</td></tr></table>	Period		EF _{CO2,y}	Start	End	MWh	12 10 2022	31 12 2022	0.923	01 01 2023	31 12 2023	0.908	01 01 2024	31 12 2024	0.908	01 01 2025	31 12 2025	0.908	01 01 2026	31 12 2026	0.908	01 01 2027	31 12 2027	0.908	01 01 2028	31 12 2028	0.908	01 01 2029	31 12 2029	0.908	01 01 2030	31 12 2030	0.908	01 01 2031	31 12 2031	0.908	01 01 2032	11 10 2032	0.908
Period		EF _{CO2,y}																																						
Start	End	MWh																																						
12 10 2022	31 12 2022	0.923																																						
01 01 2023	31 12 2023	0.908																																						
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01 01 2029	31 12 2029	0.908																																						
01 01 2030	31 12 2030	0.908																																						
01 01 2031	31 12 2031	0.908																																						
01 01 2032	11 10 2032	0.908																																						
Justification of choice of data or description of measurement methods and procedures applied	The applicable Standardized Baseline (ASB0040-2018: Grid Emission Factor for Southern African Power Pool) expired on 06/10/2021. Publicly available data was used in accordance with paragraph 23 of AMS-I.D. Which states: Calculate the weighted average emissions (in tCO ₂ /MWh) of the current generation mix.																																							
Purpose of data	Calculation of baseline emissions																																							
Comments																																								

Data / Parameter	EF _{CO2,i,y}
Data unit	tCO ₂ /tonne of biomass
Description	Default emission factor for biomass residue transportation
Source of data	Paragraph 31 of AMS-I.F

Value applied	0.0142
Justification of choice of data or description of measurement methods and procedures applied	The project falls within the small-scale definition and therefore the default emission factor for small-scale projects is applied.
Purpose of data	Calculation of project emissions
Comments	

Data / Parameter	$NCV_{i,y}$
Data unit	GJ/tonne
Description	Net calorific value of diesel fuel type I in year y
Source of data	IPCC default values at the upper limit of the uncertainty at a 95% confidence interval as provided in Table 1.2 of Chapter 1 of Vol. 2 (Energy) of the 2006 IPCC Guidelines on National GHG Inventories.
Value applied	43.3
Justification of choice of data or description of measurement methods and procedures applied	IPCC default values at the upper limit of the uncertainty at a 95% confidence interval as provided in Table 1.2 of Chapter 1 of Vol. 2 (Energy) of the 2006 IPCC Guidelines on National GHG Inventories.
Purpose of data	Calculation of project emissions
Comments	At validation only Diesel was identified as a fossil fuel to be primarily used for start-up. As such i = Diesel.

Data / Parameter	$EF_{CO_2,i,y}$
Data unit	tCO ₂ e/GJ
Description	CO ₂ emission factor for diesel fuel type i in year y
Source of data	IPCC default values at the upper limit of the uncertainty at a 95% confidence interval as provided in Table 1.4 of Chapter 1 of Vol. 2 (Energy) of the 2006 IPCC Guidelines on National GHG Inventories
Value applied	0.0726
Justification of choice of data or description of	Future revision of the IPCC Guidelines will be taken into account.

measurement methods and procedures applied	
Purpose of data	Calculation of project emissions.
Comments	At validation only Diesel was identified as a fossil fuel to be primarily used for start-up. As such i = Diesel.

Data / Parameter	EF _{CO2,LE}
Data unit	tCO ₂ /GJ
Description	CO ₂ emission factor of the most carbon intensive fuel used in the country
Source of data	South Africa Carbon tax Bill, Schedule 1, Sub-bituminous coal
Value applied	0.0691
Justification of choice of data or description of measurement methods and procedures applied	The Carbon tax bill is the most recent publication of emissions factors used nationally.
Purpose of data	Calculate leakage emissions
Comments	Although there are a few fuels in the schedule that indicate a higher emission factor they are not applied in the country.

Data / Parameter	Biomass residues categories and quantities used in the project activity
Data unit	Biomass residues category and biomass residues fate in the absence of the project activity: B1: The biomass residues are dumped or left to decay mainly under aerobic conditions. This applies, for example, to dumping and decay of biomass residues on fields; B2: The biomass residues are dumped or left to decay under clearly anaerobic conditions. This applies, for example, to landfills which are deeper than five meters. This does not apply to biomass residues that are stock-piled or left to decay on fields; B4: The biomass residues are used for energy or non-energy applications, or the primary source of the biomass residues and/or their fate cannot be clearly identified.
Description	Description of biomass categories used in the project activity, their type, source, fate in the absence of the project activity, use in the baseline scenario and quantity used.
Source of data	Provided by the PD by: (a) Type (b) Source (e.g. produced on-site, obtained from an identified biomass

Value applied	producer, obtained from biomass residues market, etc.) (c) Fate in the absence of the project activity (d) Use in the project scenario				
	Biomass residues category	Biomass residue type	Biomass residues source	Biomass residues fate in the absence of the project activity	Biomass residue use in the project scenario
	(k)			B1, B2, B3, B4	Quality (Tonnes/year)
	1	Saw Dust	On-site production	B1	Electricity generation on-site (biomass-only boiler)
	1	Saw Dust	Off-site from an identified mill	B2	Electricity generation on-site (biomass-only boiler)
	1	Saw Dust	Off-site from an identified mill	B1	Electricity generation on-site (biomass-only boiler)
	1	Saw Dust	Off-site from an unidentified mill	B4	Electricity generation on-site (biomass-only boiler)
	2	Bark	On-site production	B1	Electricity generation on-site (biomass-only boiler)
	2	Bark	On-site production	B2	Electricity generation on-site (biomass-only boiler)
	3	Pine logs	On-site production	B1	Electricity generation on-site (biomass-only boiler)
	3	Pine logs	On-site production	B2	Electricity generation on-site (biomass-only boiler)
	4	Pine chips	On-site production	B1	Electricity generation on-site (biomass-only boiler)
	4	Pine chips	On-site production	B2	Electricity generation on-site (biomass-only boiler)
	5	Gum slats	On-site production	B1	Electricity generation on-site (biomass-only boiler)
	5	Gum slats	On-site production	B2	Electricity generation on-site (biomass-only boiler)

	6	Pine slats	On-site production	B1	Electricity generation on-site (biomass-only boiler)	
	6	Pine slats	On-site production	B2	Electricity generation on-site (biomass-only boiler)	2,190
	Total					78,475
Justification of choice of data or description of measurement methods and procedures applied						
Purpose of data	Calculation of leakage emissions					
Comments						

5.2 Data and Parameters Monitored

Data / Parameter	$EG_{BL,y}$ or $EG_{PJ,facility,i,y}$
Data unit	MWh
Description	Quantity of electricity generated and supplied by the project power plant to the electricity consuming facility i in year y.
Source of data	Data will be obtained through on-site measurement.
Description of measurement methods and procedures to be applied	Gross electricity generation will be measured by the on-site electricity meter. The meter recordings will be recorded on a monthly basis. The meter will be maintained and calibrated in accordance with the recommendations of the supplier.
Frequency of monitoring/recording	Continuous measurement and recorded monthly

Value applied	<table><tr><th colspan="2">Period</th><th>EG_{BL,y}</th></tr><tr><th>Start</th><th>End</th><th>MWh</th></tr><tr><td>12 10 2022</td><td>31 12 2022</td><td>4,485</td></tr><tr><td>01 01 2023</td><td>31 12 2023</td><td>20,463</td></tr><tr><td>01 01 2024</td><td>31 12 2024</td><td>20,463</td></tr><tr><td>01 01 2025</td><td>31 12 2025</td><td>20,463</td></tr><tr><td>01 01 2026</td><td>31 12 2026</td><td>20,463</td></tr><tr><td>01 01 2027</td><td>31 12 2027</td><td>20,463</td></tr><tr><td>01 01 2028</td><td>31 12 2028</td><td>20,463</td></tr><tr><td>01 01 2029</td><td>31 12 2029</td><td>20,463</td></tr><tr><td>01 01 2030</td><td>31 12 2030</td><td>20,463</td></tr><tr><td>01 01 2031</td><td>31 12 2031</td><td>20,463</td></tr><tr><td>01 01 2032</td><td>11 10 2032</td><td>15,866</td></tr></table>	Period		EG _{BL,y}	Start	End	MWh	12 10 2022	31 12 2022	4,485	01 01 2023	31 12 2023	20,463	01 01 2024	31 12 2024	20,463	01 01 2025	31 12 2025	20,463	01 01 2026	31 12 2026	20,463	01 01 2027	31 12 2027	20,463	01 01 2028	31 12 2028	20,463	01 01 2029	31 12 2029	20,463	01 01 2030	31 12 2030	20,463	01 01 2031	31 12 2031	20,463	01 01 2032	11 10 2032	15,866
Period		EG _{BL,y}																																						
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01 01 2030	31 12 2030	20,463																																						
01 01 2031	31 12 2031	20,463																																						
01 01 2032	11 10 2032	15,866																																						
Monitoring equipment	Electricity meter																																							
QA/QC procedures to be applied	The electricity meter will be subject to regular maintenance and testing in accordance with the stipulation of the meter supplier or national requirements.																																							
Purpose of data	Calculation of baseline emissions																																							
Calculation method	N/A																																							
Comments																																								

Data / Parameter	FR _{f,m}
Data unit	tonnes
Description	Total mass of freight transported in freight transportation activity <i>f</i> in monitoring period <i>m</i>
Source of data	Records of vehicle operator or records obtained from a identified biomass residue producer. For the PD the hourly biomass consumption design parameter (9 t/hour) was applied.
Description of measurement methods and procedures to be applied	Use weighbridge or scale. Data monitored continuously (per load/truck arriving at the weighbridge or transported to boiler) and aggregated as appropriate.
Frequency of monitoring/recording	Continuously monitored and recorded annually

Value applied	Periods		Total biomass residue transported ($FR_{t,m}$)
	Start	End	
	12 10 2022	31 12 2022	17,200
	01 01 2023	31 12 2023	78,475
	01 01 2024	31 12 2024	78,475
	01 01 2025	31 12 2025	78,475
	01 01 2026	31 12 2026	78,475
	01 01 2027	31 12 2027	78,475
	01 01 2028	31 12 2028	78,475
	01 01 2029	31 12 2029	78,475
	01 01 2030	31 12 2030	78,475
	01 01 2031	31 12 2031	78,475
	01 01 2032	11 10 2032	60,845
Monitoring equipment	A weighbridge or weight-scale(s)		
QA/QC procedures to be applied	-		
Purpose of data	Calculation of project emissions		
Calculation method	The weight of full trucks is measured on the way in from this the weight of the empty truck is weighed on the way out.		
Comments	-		

Data / Parameter	$FC_{i,j,y}$
Data unit	tonne/year
Description	Quantity of diesel fuel type i combusted in process j during the year y.
Source of data	Onsite measurement
Description of measurement methods and procedures to be applied	The diesel utilized is sourced from an onsite storage facility. The volume used will be measured using a volume meter and recorded monthly. For diesel brought onto the site, the invoices will be used record the onsite diesel consumption.
Frequency of monitoring/recording	Continuously monitored and recorded monthly

Value applied	Recording periods		Measured value	Measured value
	Start	End	Start-up fuel (t)	Biomass Processing (t)
	12 10 2022	31 12 2022	15.78	1.58
	01 01 2023	31 12 2023	72.00	7.20
	01 01 2024	31 12 2024	72.00	7.20
	01 01 2025	31 12 2025	72.00	7.20
	01 01 2026	31 12 2026	72.00	7.20
	01 01 2027	31 12 2027	72.00	7.20
	01 01 2028	31 12 2028	72.00	7.20
	01 01 2029	31 12 2029	72.00	7.20
	01 01 2030	31 12 2030	72.00	7.20
	01 01 2031	31 12 2031	72.00	7.20
	01 01 2032	11 10 2032	55.82	5.58
Monitoring equipment	Flow meter(s), depth stick(s)/rulers			
QA/QC procedures to be applied	Metered fuel consumption will be cross-checked with purchased fuel invoices from the financial records.			
Purpose of data	Calculation of project emissions			
Calculation method	Invoices will be used to establish the quantity of fuel used and the fractions for start-up and processing established.			
Comments	In the PD for validation Start-up diesel consumption is assessed to be 6 tonne/month and Process diesel consumption at 0.6 tonne/month.			

Data / Parameter	$BR_{PJ,n,y}$
Data unit	tonnes on a dry basis
Description	Quantity of biomass residues of category n used in facilities which are located at the project site and included in the project boundary in year y
Source of data	On-site measurements or data provided by an identified biomass residue producer inside the project boundary.
Description of measurement methods and procedures to be applied	Use weight meters. Adjust by the moisture content in order to determine the quantity of dry biomass.

Frequency of monitoring/recording	Data monitored continuously and aggregated as appropriate, to calculate emissions reductions																																																																				
Value applied	<table border="1"> <thead> <tr> <th>Biomass residues category</th><th>Biomass residue category name</th><th>B1</th><th>B2</th><th>B4</th><th>Moisture content of the biomass residues</th><th>Biomass residues quantity</th></tr> <tr> <th>(k)</th><th></th><th>Tonnes/year</th><th>Tonnes/year</th><th>Tonnes/year</th><th>(%)</th><th>(dry-tonnes)</th></tr> </thead> <tbody> <tr> <td>1</td><td>Saw Dust</td><td>-</td><td>68,620.00</td><td>-</td><td>45.0%</td><td>37,741.00</td></tr> <tr> <td>2</td><td>Bark</td><td>-</td><td>730.00</td><td></td><td>49.0%</td><td>372.30</td></tr> <tr> <td>3</td><td>Pine logs</td><td>-</td><td>1,460.00</td><td></td><td>37.6%</td><td>911.04</td></tr> <tr> <td>4</td><td>Pine chips</td><td>-</td><td>3,285.00</td><td></td><td>31.2%</td><td>2,260.08</td></tr> <tr> <td>5</td><td>Gum slats</td><td>-</td><td>2,190.00</td><td></td><td>31.2%</td><td>1,506.72</td></tr> <tr> <td>6</td><td>Pine slats</td><td>-</td><td>2,190.00</td><td></td><td>37.1%</td><td>1,377.51</td></tr> <tr> <td colspan="2">Total</td><td>-</td><td>78,475.00</td><td>-</td><td></td><td>44,168.65</td></tr> </tbody> </table>						Biomass residues category	Biomass residue category name	B1	B2	B4	Moisture content of the biomass residues	Biomass residues quantity	(k)		Tonnes/year	Tonnes/year	Tonnes/year	(%)	(dry-tonnes)	1	Saw Dust	-	68,620.00	-	45.0%	37,741.00	2	Bark	-	730.00		49.0%	372.30	3	Pine logs	-	1,460.00		37.6%	911.04	4	Pine chips	-	3,285.00		31.2%	2,260.08	5	Gum slats	-	2,190.00		31.2%	1,506.72	6	Pine slats	-	2,190.00		37.1%	1,377.51	Total		-	78,475.00	-		44,168.65
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Monitoring equipment	Weighbridge and/or conveyor load cell(s).																																																																				
QA/QC procedures to be applied	Cross-check the measurements with an annual energy balance that is based on purchased quantities and stock changes.																																																																				
Purpose of data	Calculation of leakage emissions																																																																				
Calculation method	-																																																																				
Comments																																																																					

Data / Parameter	NCV _{n,y}
Data unit	GJ/tonnes on dry-basis
Description	Net calorific value of biomass residues of category n in year y
Source of data	On-site measurements or data provided by an identified biomass residue producer inside the project boundary.
Description of measurement methods and procedures to be applied	Measurements shall be carried out at reputed laboratories and according to relevant international standards. Measure the NCV on dry-basis
Frequency of monitoring/recording	At least every six months, taking at least three samples for each measurement

Value applied			
	Biomass residues category	Biomass residue category name	NCV _{n,y}
	(k)		(GJ/tonne)
	1	Saw Dust	11.59
	2	Bark	7.32
	3	Pine logs	10.33
	4	Pine chips	8.69
	5	Gum slats	11.59
	6	Pine slats	11.68
Monitoring equipment	Samples will be taken at the weighbridge and/or at the conveyer loadcell(s) and analyses at the in-house lab.		
QA/QC procedures to be applied	Check the consistency of the measurements by comparing the measurement results with measurements from previous years, relevant data sources (e.g. values in the literature, values used in the national GHG inventory) and default values by the IPCC. If the measurement results differ significantly from previous measurements or other relevant data sources, additional measurements are conducted. Ensure that the NCV is determined on the basis of dry biomass.		
Purpose of data	Calculation of emission reductions		
Calculation method	N/A		
Comments	In the case of missing data, the following can be used for project activities, where end users of the subsystems or measures are households/communities/small and medium enterprises (SMEs): - IPCC default values at the upper limit of the uncertainty at a 95 percent confidence interval as provided in table 1.2 of Chapter 1 of Vol.2 (Energy) of the 2006 IPCC Guidelines on National GHG Inventories; or - The highest value from previous monitoring periods of the same biomass type		

Data / Parameter	Moisture content of the biomass residues																							
Data unit	% Water content in mass basis in wet biomass residues																							
Description	Moisture content of each biomass residues type n																							
Source of data	On-site measurements or data provided by an identified biomass residue producer inside the project boundary.																							
Description of measurement methods and procedures to be applied	The moisture content per biomass residue category (k) will be measured and/or derived from samples collected at the weighbridge or at the conveyer load cell(s) or both.																							
Frequency of monitoring/recording	The moisture content is monitored for each batch of biomass of homogeneous quality. The weighted average is calculated for each monitoring period and used in the calculations.																							
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Monitoring equipment	Moisture content will be measured at the weighbridge and/or at the conveyer loadcell(s) or samples will be taken at the same places and send for analyses to a third party lab.																							
QA/QC procedures to be applied	-																							
Purpose of data	Calculation of emission reductions																							
Calculation method	-																							
Comments	In the absence of measurements per load of biomass residue category the stockpile average will be used for that period for both biomass residue under scenario B1, B2 and B4. In the PD for validation moisture content estimates were applied.																							

5.3 Monitoring Plan

The monitoring plan will be implemented by the project proponents (FX Group (Pty) Ltd. and Brundtland (Pty) Ltd.). FX Group (Pty) Ltd will be responsible for the collection of data, which will be quality reviewed by Brundtland (Pty) Ltd. The monitoring plan has been designed to ensure that the variables monitored to determine the emission reductions realised by the project will be monitored transparently and clearly during the crediting period.

1. Monitoring Approach

The project proponent has made an employee of FX Group (Pty) Ltd responsible for the data collection and handling (termed the 'data manager') as described in section 5.1 and 5.2 of the PD and outlined in the monitoring diagram in section 3.3. The data manager will collect and aggregate the data from each of the metering points (see section 3.3) on a monthly basis and provide it to the responsible person at Brundtland (Pty) Ltd. for review and reporting purposes. Following review, any discrepancies will be highlighted and communicated back to the data manager for amendment or clarification.

2. QA/QC and back-up procedure

In any case where problems occur which can affect the quality of the data, the data manager will initiate and supervise the implementation of corrective actions. In order to avoid any kind of discrepancies in the monitoring procedures in cases of potential breakdowns, the following emergency preparedness plan will be executed:

- The centralised data monitoring facilities at the project control room will be considered as the primary source from which on-site monitored data is captured/stored and backed up. All data streams are archived both digitally and physically, providing back-up in the case of data loss or other errors;
- If some or all of these data streams malfunction or show discrepancies, data from the individual meter at the defined monitoring point will be applied as a substitute;
- In instances where individual meters malfunction, alternative methods will be applied by the plant operators (e.g. mobile/handheld meters, etc.).

What exact alternative approach is applied, will be clearly captured in the monthly data collection sheets. Mechanical and electrical preventative check sheets are conducted on a weekly basis to prevent unnecessary breakdowns within the power plant. Continuous water sampling is done to prevent unnecessary breakdown on Power Plant boiler tubes, which in turn will increase unnecessary stop and start of Power Plant.

The diesel used for plant start-up and for front-end loaders is recorded through a diesel flow meter. Diesel usage for each usage type is recorded daily and archived digitally. The total electricity generated by the plant is recorded using an electricity meter, measured directly from DCS (Distributed Control System). Records of power generation is back traceable on the DCS itself. Data collection for the power plant is conducted hourly and aggregated both daily and monthly. The conveyor weight-scale measures the weight of biomass consumed by the plant continuously. All the data from the weight-scale is back traceable on the DCS and is recorded digitally.

3. Meters

The data manager ensures that all meters applied are maintained and calibrated in accordance with the manufacturer recommendations and, where relevant, national legislation or standards. In the event that a meter is reaching the end of its calibration period, the process for re-calibration will be coordinated by the data manager and representatives from Brundtland (Pty) Ltd. This is to ensure that each meter is correctly calibrated and that the data collection method and process during re-calibration is clearly defined and followed.

4. Biomass flow and measurement points

As presented in the system diagram, the biomass arrives from saw-mills at the weighbridge, where the weight and source of each truck of biomass is recorded, both physically (through weighbridge tickets) and digitally. Following this, the biomass destined for the power plant is fed through a chipper and stored at the project stock-pile. Prior to entering the power plant, the weight of biomass being fed into the plant is recorded on a conveyor belt weight-scale, such that all biomass fed into the boiler is accounted for.

APPENDIX 1: COMMERCIALY SENSITIVE INFORMATION

Section	Information	Justification

APPENDIX X: <TITLE OF APPENDIX>